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Widely targeted metabolomics analysis and antioxidant properties of *Camellia oleifera* seeds of different cultivars

Zuoyi Zhu [a], Jingfang Lv [b], Jiarong Yao [a], Zhen Li [a], Shuai Yang [a], Xueping Zhao [a,*], Fen Dai [a,*]

[a] Institute of Agro-product Safety and Nutrition, Zhejiang Academy of Agricultural Sciences, 310021 Hangzhou, China
[b] Shangyu Inspection and Test Center for Agricultural Products Quality and Safety, 312300 Shaoxing, China

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ABSTRACT

Camellia oleifera has attracted significant attention due to its seed oil. However, there is limited information available on the metabolites of *Camellia oleifera* seeds. In this study, we compared the seed metabolites of four *Camellia oleifera* cultivars (CL4, CL18, CL40, and CL53) using a UPLC-MS/MS-based widely targeted metabolomics approach. A total of 1339 metabolites, categorized into 13 distinct classes, were identified. Comparative analysis revealed that there were obvious differences in the metabolic profiles of *Camellia oleifera* seeds, with 330 metabolites showing significantly differential accumulation among four cultivars. Notably, substantial variations were observed in the accumulation of flavonoids and phenolic acids. Metabolic pathway analysis of differential metabolites highlighted significant enrichment in the "biosynthesis of quercetin aglycones II" and "flavonoid biosynthesis" pathways. Antioxidant assays demonstrated that CL53 seed samples exhibited the highest antioxidant capacity. Overall, this study provided scientific support for *Camellia oleifera* cultivar selection and high-value utilization of *Camellia oleifera* seeds.

1. Introduction

Camellia oleifera Abel., a plant belonging to the Camellia genus of the Theaceae family, is widely cultivated in southern China, particularly in the provinces of Hunan, Jiangxi, Zhejiang, and Guangxi (Shi et al., 2020; Zhang et al., 2022). As one of the most valuable woody oleiferous species, *Camellia oleifera* has been cultivated in China for over 2300 years (Lin et al., 2018). China is both the origin and the largest producer of *Camellia oleifera* globally, accounting for 90 % of the world's production (Gao et al., 2024). The seeds of *Camellia oleifera*, with an oil content of approximately 40 %, are recognized as one of the four major woody oil crops worldwide, alongside oil palm, coconut, and olive crops.

Camellia oleifera seed oil is a high-quality edible oil and has garnered significant attention due to its high content of unsaturated fatty acids (UFAs), which constitute approximately 90 % of its total fatty acids (FAs). With FAs profile remarkably similar to that of olive oil, *Camellia oleifera* seed oil is often referred to as the "oriental olive oil". The high levels of oleic acid (C18:1, 76.1–83.3 %) and linoleic acid (C18:2, 7.0–11.0 %) in *Camellia oleifera* seed oil contribute to its exceptional nutritional value and health benefits (Gao et al., 2024; Luan et al., 2020). In recent years, the Food and Agriculture Organization has recommended *Camellia oleifera* seed oil as a healthy cooking oil (Wang et al., 2017). Beyond its use as an edible oil, *Camellia oleifera* seed oil is also widely utilized in the fields of functional foods, cosmetics and pharmaceuticals.

In addition to UFAs, such as oleic acid and linoleic acid, *Camellia oleifera* seeds are rich in a variety of bioactive compounds with health-promoting properties, such as sterols, squalene, polyphenols and tocopherols (Zhang et al., 2022). These components endow *Camellia oleifera* seed oil with diverse pharmacological effects and biological activities, such as hypolipidemic, neuroprotective, skin repair, osteological modifying, anti-obesity, antioxidant, anti-inflammatory, antimicrobial, hepatoprotective, gastrointestinal, anti-tumor, and melanogenesis activities (Gao et al., 2024). Currently, most research has focused on analyzing and quantifying the fatty acid composition, squalene and tocopherols compounds in *Camellia oleifera* seeds, while other potentially bioactive compounds have received limited attention. Furthermore, the levels of these bioactive substances in *Camellia oleifera* seeds are significantly influenced by factors such as cultivar, growing environment, and cultivation techniques.

In recent years, widely targeted metabolomic analysis, primarily based on UPLC-MS/MS, has emerged as a powerful approach. This

* Corresponding authors.
E-mail addresses: zhaoxp@zaas.ac.cn (X. Zhao), daif@zaas.ac.cn (F. Dai).

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method integrates the advantages of both non-targeted and targeted metabolomics and has been extensively applied in the analysis of plant metabolites across various crops, fruits, and tea (Hou et al., 2023; Li et al., 2022; Wang et al., 2023). Several studies have explored the nutrient composition, flavor profiles, and health benefits of *Camellia oleifera* seed oil (Gao et al., 2024; Zhang et al., 2022). Luan et al. (2020) isolated and structurally identified 249 chemical constituents, including triterpenoid saponins, flavonoids, polyphenols and polysaccharides, from different parts of *Camellia oleifera*, such as seed, root, stem, and flower bud. Song et al. (2022) elucidated the composition and distribution of flavonoids and molecular mechanism of flavonoid biosynthesis in the *Camellia oleifera* seeds. Hong et al. (2019) identified the polyphenols in *Camellia oleifera* seed cakes by UPLC-Q-TOF-MS/MS. Despite these advancements, there remains a lack of comprehensive studies on the chemical composition and nutritional properties of *Camellia oleifera* seeds. With the application of widely targeted metabolomics, it is now feasible to perform qualitative and quantitative analysis of thousands of metabolites in a single assay, making it an effective approach for studying the metabolites of *Camellia oleifera* seeds and comparing their metabolite profiles across different cultivars.

To the best of our knowledge, there is limited information available on the application of metabolomics as a tool for comprehensively profiling metabolites in *Camellia oleifera* seeds. In this study, *Camellia oleifera* seeds of four representative cultivars were selected for analysis. A widely targeted metabolomics approach was employed to investigate the types and abundance of primary and specialized metabolites in the seeds of these four cultivars. Additionally, a comparative analysis was conducted to identify significantly differential metabolites and functional components associated with antioxidant activities among the cultivars. The findings of this study will provide a theoretical foundation for the future utilization of *Camellia oleifera* seeds in the food industry.

2. Materials and methods

2.1. Plant materials

Camellia oleifera cultivars ‘changlin 4’ (CL4), ‘changlin 18’ (CL18), ‘changlin 40’ (CL40) and ‘changlin 53’ (CL53), which were grown in the *Camellia oleifera* planting base, Changshan, Zhejiang Province, China (28°9′N, 118°6′E), were selected in this study. Fourteen-year-old *Camellia oleifera* trees with good growth and similar growth potential were randomly selected. For each cultivar, mature fruits were randomly collected from six trees in October. Three biological replicates of each cultivar with two trees per replicate were included in the experiment. The transverse diameter, longitudinal diameter and weight of the fresh fruits were measured. The collected fruits were shelled to obtain seeds, and then the seeds were shelled to obtain kernels on the ice. The pericarp thickness, seed yield and kernel yield of fruits from different *Camellia oleifera* cultivars were measured. For each sample, part of the kernels were stored in a freezer at −80 °C for metabolomics analysis, while the others were dried at 50 °C for and ground into powder for analysis of common nutritional components and antioxidant activity.

2.2. Chemicals and reagents

2,2′-azino-bis(3-ethyl-benzothiazoline-6-sulfonic acid) diammonium salt (ABTS), 2,4,6-tri(2-pyridyl)-*s*-triazine (TPTZ), 2,2′-diphenyl-1-picrylhydrazyl (DPPH), 6-hydroxy-2,5,7,8-tetramethylchromane-2-carboxylic acid (Trolox) and a Supelco mixed standard containing 37 fatty acid methyl ester (FAME. C4−C24) were purchased from Sigma-Aldrich Chemical Co., Ltd. (St. Louis, MO, USA). Folin-Ciocalteu's phenol reagent, gallic acid and rutin were purchased from Macklin Biochemical Co., Ltd. (Shanghai, China). Methanol, formic acid and *n*-hexane were of HPLC-grade and purchased from the Merck Group (Darmstadt, Germany). All other reagents used were of analytical grade and purchased from Shanghai Yuanye Biotechnology Co., Ltd. (Shanghai, China).

2.3. Determination of oil content, protein content, TPC, TFC and fatty acid composition

The oil content (%) and protein content (%) were measured according to the national standard GB 5009.6-2016 and GB 5009.5-2016 of China, respectively. Total phenolic content (TPC, expressed as milligrams of gallic acid equivalent (GAE) per gram of sample) was assessed using the Folin-Ciocalteu method (Gan et al., 2017). Total flavonoid content (TFC, expressed as milligrams of rutin equivalent (RE) per gram of sample) was determined using the aluminum trichloride colorimetric method according to a previous reference (Shraim et al., 2021). The fatty acid composition (%) was determined according to the national standard (GB 5009.168-2016). Qualitative analysis of fatty acids was based on retention time of FAME and quantitative analysis was based on the percentage of relative peak areas.

2.4. Sample preparation and extraction for widely targeted metabolomics analysis

The *Camellia oleifera* kernels were vacuum freeze-dried and then ground into powder. Then, 50 mg powder of each kernel sample was dissolved in 1.2 mL of 70 % aqueous methanol and vortexed. The extracts were centrifuged at 14,000 × g for 10 min, and then the supernatant was collected and filtered through a microporous membrane (0.22 μm) before UPLC-MS/MS analysis. Quality control (QC) samples were prepared by mixing the sample extracts.

2.5. Widely targeted metabolomics analysis

Metabolomic analysis was performed using an UPLC-ESI-MS/MS system (UPLC, SHIMADZU Nexera X2; MS, Applied Biosystems 4500 Q TRAP) as previously described (Li et al., 2022). Metabolite identification was performed by searching the self-constructed internal database MWDB (MetWare Biotechnology Co., Ltd., Wuhan, China) and public databases, such as MassBank (http://www.massbank.jp), KNApSAcK (http://kanaya.naist.jp/KNApSAcK), HMDB (http://www.hmdb.ca), and METLIN (http://metlin.scripps.edu/index.php). The metabolites were identified by comparing their retention time, accurate precursor ion (Q1), product ion (Q3) values and fragmentation pattern with the databases. Metabolite quantification was carried out via the MRM mode of QqQ mass spectrometry. In the MRM mode, the quadrupole firstly searched for precursor ions of target substances. The precursor ions were then fragmented to form fragment ions, which were filtered through QqQ to select single fragment ions with desired characteristics, while eliminating the interference from non-target ions. This step leads to high precision and repeatability of quantification results. After metabolite mass spectrometry data were obtained for different samples, all mass spectrum peaks were subjected to area integration and integration correction. The relative contents of metabolite in each sample were represented with peak area integrals.

2.6. Antioxidant activity assays

Sample preparation: the kernel powder (0.5 g) was added with 100 mL of 70 % aqueous methanol and sonicated for 30 min, followed by centrifugation at 8000 × g for 10 min. The resulting supernatant was collected for antioxidant activity assays.

2.6.1. ABTS radical scavenging activity

The ABTS radical scavenging activity (RSA) was assessed according to a previous method (Herrera-Balandrano et al., 2021). The sample supernatant was mixed with ABTS mixture solution and then incubated in the dark for 10 min. Subsequently, the absorbance (A) at 734 nm was measured. The blank control was a mixture of the solvent and ABTS solution. The ABTS RSA rates were calculated as follows.

$$ABTSRSA(\%) = \left(A_{control} - A_{sample}\right)/A_{control} \times 100 \quad (1)$$

where A_{sample} denotes the absorbance of the sample extract and $A_{control}$ denotes the absorbance of the blank control.

2.6.2. DPPH radical scavenging activity

The DPPH RSA was assessed according to a previous method (Lee et al., 2022). The sample supernatant was mixed with DPPH solution and then incubated in the dark at 30 °C for 20 min. The absorbance (A) was measured at 517 nm. The following equation was used to determine the DPPH RSA rates of the samples.

$$DPPHRSA(\%) = \left(A_{control} - A_{sample}\right)/A_{control} \times 100 \quad (2)$$

where A_{sample} denotes the absorbance of the sample extract and $A_{control}$ denotes the absorbance of the blank control.

2.6.3. Hydroxyl radical scavenging activity

The hydroxyl RSA was assessed according to a previously described method (Zhu et al., 2024). In brief, the same volume of $FeSO_4$ solution (6 mmol/L), salicylic acid (9 mmol/L in ethanol) and *Camellia oleifera* seed extract were mixed. After shaking well, the same volume of H_2O_2 solution (20 mmol/L) was added. Finally, the absorbance at 510 nm was measured after incubating at 37 °C for 30 min. The following equation was used to determine the hydroxyl radical RSA rates of the samples.

$$HydroxylRSA(\%) = \left(A_{control} - A_{sample}\right)/A_{control} \times 100 \quad (3)$$

where A_{sample} denotes the absorbance of the sample extract and $A_{control}$ denotes the absorbance of the blank control.

2.6.4. Ferric-reducing antioxidant power (FRAP) test

A previous method was used to assess the FRAP of the samples (Grzesik et al., 2018). The sample supernatant was mixed with the FRAP working solution, followed by incubating at 37 °C for 20 min. The absorbance was measured at 593 nm. The reducing power was calculated and expressed as micromoles of trolox equivalent per gram of dry weight (μmol Trolox/g DW).

2.7. Statistical analysis

All measurements were repeated in triplicate and expressed as mean ±standard deviation. Significance was assessed by one way analysis of variance (ANOVA) combined with Duncan's test using SPSS 20.0 (SPSS Inc., Chicago, IL, USA). Benjamini-Hochberg method was employed to control the false discovery rate (FDR) (Benjamini & Hochberg, 1995). SIMCA-P software was used to analyze the UPLC–MS/MS data via principal component analysis (PCA), orthogonal partial least squares-discriminant analysis (OPLS-DA) and hierarchical cluster analysis (HCA). Furthermore, differential metabolites were identified with the variables of p-value < 0.05, FDR < 0.05, variable importance in the OPLS-DA projection (VIP > 1.0) and fold changes (FC) > 2 or < 0.5.

2.8. Kyoto encyclopaedia of genes and genomes (KEGG) annotations and enrichment analysis

Identified metabolites were annotated using KEGG Compound database, and annotated metabolites were then mapped to KEGG Pathway database. The mapped pathways were subjected to metabolite sets enrichment analysis. The significance was considered with the variables of p-value < 0.05.

3. Results

3.1. Phenotypic differences among Camellia oleifera cultivars

The four Camellia oleifera cultivars (CL4, CL18, CL40, and CL53) exhibited distinct morphological variations in fruit characteristics despite being cultivated under identical seasonal and environmental conditions (Fig. 1). Notable differences were observed in both coloration and shape parameters. The fruits of CL4 and CL18 displayed a green base coloration overlaid by reddish pigmentation, whereas CL40 and CL53 presented uniformly green fruits with distinctive speckling patterns. Quantitative analysis of morphological parameters revealed CL53 as the cultivar with superior dimensional traits, demonstrating significantly greater values in transverse diameter (width), longitudinal diameter (height), and whole-fruit weight compared with the other cultivars (Table 1). CL4 produced characteristically oval-shaped fruits, while the other three cultivars (CL18, CL40, and CL53) developed fruits with predominantly spherical to oblate-spherical morphologies.

3.2. Economic characteristics and nutritional contents

Quantitative analysis of seed and kernel yields revealed significant differences among the four cultivars (Table 1). CL18 exhibited the highest seed yield (51.92 ± 1.43 %) and kernel yield (59.40 ± 0.51 %), whereas CL53 showed the lowest values for both parameters (seed yield: 43.93 ± 0.88 %; kernel yield: 51.64 ± 0.61 %). The oil content across cultivars ranged from 34.41 ± 0.85 % to 39.85 ± 1.58 %, with CL53 displaying a significantly lower oil content compared with CL4, CL18, and CL40 ($p < 0.05$). The protein contents varied among the four different cultivars and ranged from 6.22±0.04 % to 8.09±0.24 % (Table 2). CL4 (6.22 ± 0.04 %) and CL40 (6.97 ± 0.05 %) exhibited relatively low protein levels, while CL18 (7.90 ± 0.42 %) and CL53 (8.09 ± 0.24 %) demonstrated higher contents. Regarding polyphenol, TPC of *Camellia oleifera* seeds of four cultivars ranged from 23.92±0.71 mg of GAE/g to 37.33±0.61 mg of GAE/g, with CL53 showing the highest TPC, followed by CL4, CL40 and CL18. TFC displayed a similar tendency across cultivars as TPC.

The composition of FAs in the *Camellia oleifera* seed oil was investigated, and six kinds of FAs were identified. The six FAs included 3 saturated FAs (palmitic, stearic and arachidic acids, 16:0, 18:0 and 20:0), 1 monounsaturated FAs (oleic acid, 18:1) and 2 polyunsaturated FAs (linoleic and linolenic acids, 18:2 and 18:3). UFAs dominated the profile, collectively accounting for approximately 90 % of the total FAs, with oleic acid (18:1) representing approximately 80 %. Among saturated FAs, palmitic acid (16:0) was the most abundant. For polyunsaturated FAs, linoleic acid (18:2) predominated over linolenic acid (18:3). Notably, CL18 diverged from other cultivars in its FAs composition, exhibiting lower oleic acid (18:1) levels and higher linoleic acid (18:2) content. Apart from this, the overall FAs profiles remained broadly consistent across the four cultivars. To assess the nutritional values of FAs, polyunsaturated fatty acid/saturated fatty acid ratio (PUFA/SFA), index of atherogenicity (IA), and index of thrombogenicity (IT) are the most commonly used indices as they outline significant implications and provide clear evidence. The PUFA/SFA, IA and IT values of *Camellia oleifera* seed oil obtained in this study ranged from 0.69–1.31, 0.07–0.08, and 0.18–0.22, respectively, which were consistent with the values of plant oil as reported by Chen & Liu (2020).

3.3. Metabolomic profiling of Camellia oleifera seeds

To comprehensively evaluate the phytochemical substances of *Camellia oleifera* seeds across different cultivars, widely targeted metabolomics, based on UPLC-MS/MS, was employed in this study. Metabolite identification was achieved by matching experimental data to a self-constructed plant-specific metabolite database. Total ion current (TIC) plots and multi-peak detection plots of the QC samples were



Fig. 1. *Camellia oleifera* mature fruits (upper part of the image), seeds (middle part of the image) and kernels (lower part of the image) of four cultivars. (A) CL4; (B) CL18; (C) CL40; (D) CL53.

Table 1
Economic characteristics of *Camellia oleifera* fruits of different cultivars.

Cultivar	Transverse diameter (mm)	Longitudinal diameter (mm)	Pericarp thickness (mm)	Fresh fruit weight (g)	Seed yield (%)	Kernel yield (%)	Oil content (%)
CL4	37.03±0.87^{b}	38.79±2.28^{ab}	3.21±0.39^{c}	24.70±2.20^{b}	47.71±0.56^{b}	59.15±0.58^{a}	38.41±0.70^{a}
CL18	38.04±2.20^{b}	31.06±2.31^{c}	3.17±0.39^{c}	27.17±1.53^{b}	51.92±1.43^{a}	59.40±0.51^{a}	38.29±0.61^{a}
CL40	36.05±1.65^{b}	37.09±2.22^{b}	3.71±0.47^{b}	25.63±1.42^{b}	44.27±1.29^{c}	55.94±0.42^{b}	39.85±1.58^{a}
CL53	48.28±2.12^{a}	42.89±3.80^{a}	4.43±0.44^{a}	43.73±2.46^{a}	43.93±0.88^{c}	51.64±0.61^{c}	34.41±0.85^{b}

Values represent the mean ± SD. a-c: Different letters indicated significant difference ($p < 0.05$) at the same column.

Table 2
Nutritional indicators of *Camellia oleifera* seeds of different cultivars.

Cultivar	Protein (%)	TPC (mg of GAE/g)	TFC (mg of RE/g)	Composition of fatty acid (%)							
				C16:0	C18:0	C18:1	C18:2	C18:3	C20:0	SFA	UFA
CL4	6.22± 0.04^{c}	23.92±0.71^{d}	18.30±0.73^{c}	7.29 ±0.14^{a}	2.52 ±0.09^{a}	82.61 ±0.97^{a}	6.88± 0.26^{b}	0.23 ±0.01^{c}	0.46 ±0.02^{b}	10.27 ±0.18^{a}	89.73 ±0.72^{c}
CL18	7.90± 0.42^{a}	28.83±0.24^{b}	21.92±2.30^{b}	6.70 ±0.07^{b}	1.66 ±0.01^{c}	79.59 ±0.46^{b}	11.21 ±0.13^{a}	0.38 ±0.01^{a}	0.46 ±0.01^{b}	8.83± 0.12^{c}	91.17 ±0.52^{a}
CL40	6.97± 0.05^{b}	26.30±0.82^{c}	20.24±1.02^{c}	6.20 ±0.09^{c}	2.27 ±0.12^{b}	83.77 ±0.10^{a}	7.07± 0.05^{b}	0.24 ±0.01^{c}	0.46 ±0.01^{b}	8.92± 0.15^{c}	91.08 ±0.22^{a}
CL53	8.09± 0.24^{a}	37.33±0.61^{a}	28.91±0.64^{a}	6.79 ±0.08^{b}	2.44 ±0.01^{a}	82.59 ±0.73^{a}	7.29± 0.34^{b}	0.34 ±0.01^{b}	0.55 ±0.01^{a}	9.78± 0.11^{b}	90.22 ±0.82^{b}

Values represent the mean ± SD. a-d: Different letters indicated significant difference ($p < 0.05$) at the same column.

shown in Fig. S1. Based on the metabolite database, qualitative and quantitative analyses were conducted on the metabolites in the samples. In total, 1339 metabolites were identified, including 319 flavonoids, 172 phenolic acids, 159 amino acids and derivatives, 118 lipids, 88 alkaloids, 78 lignans and coumarins, 71 terpenoids, 67 nucleotides and derivatives, 62 organic acids, 14 quinones, 12 tannins, 2 steroids and 177 other metabolites (Fig. 2A, Table S1). Notably, flavonoids, phenolic acids, amino acids, and lipids constituted the most abundant metabolite groups, while quinones and tannins were the least represented. High correlation coefficients (r = 0.94–0.97) were observed between QC samples (Fig. S2), indicating good repeatability and instrumental stability across multiple runs. Furthermore, high correlation coefficients (r = 0.93–0.98) among triplicate samples (Fig. S3) demonstrated robust sample homogeneity and experimental consistency.

3.4. Multivariate analysis of identified metabolites in Camellia oleifera seeds

To assess the overall metabolite differences among the four *Camellia oleifera* cultivars, PCA was conducted under an unsupervised model. The PCA score plot revealed clear separation among the cultivars, with the first two principal components (PC1 and PC2) accounting for 34.8 % and 19.7 % of the total variance, respectively (Fig. 2B). The distinct clustering indicated significant metabolic differences between the cultivars. Furthermore, the tight grouping of the three biological replicates for each cultivar demonstrated high reproducibility and reliability of the experimental data.

Subsequently, OPLS-DA, a supervised multivariate data analysis method, was employed to further comprehensively assess the relationship between metabolite composition and sample classification. The OPLS-DA score plot demonstrated clear separation of *Camellia oleifera* seed samples across cultivars (Fig. 2C), highlighting significant differences in their metabolic phenotypes. All samples were well-contained within the 95 % confidence intervals, as delineated by Hotelling's T-squared ellipse, further validating the robustness of the model and the distinct metabolic profiles of each cultivar.

3.5. Differential metabolites screening among different Camellia oleifera cultivars

A total of 330 significantly differential metabolites were identified across the four cultivars, including 63 flavonoids, 55 phenolic acids, 40 amino acids and derivatives, 32 lignans and coumarins, 24 lipids, 24 terpenoids, 20 alkaloids, 15 organic acids, 13 nucleotides and derivatives, 1 tannin, 1 steroid, and 42 other metabolites (Table S1). The hierarchical cluster analysis of significantly differential metabolites revealed distinct patterns of metabolite variation among cultivars (Fig. S4). The HCA results clearly segregated the seed samples into four

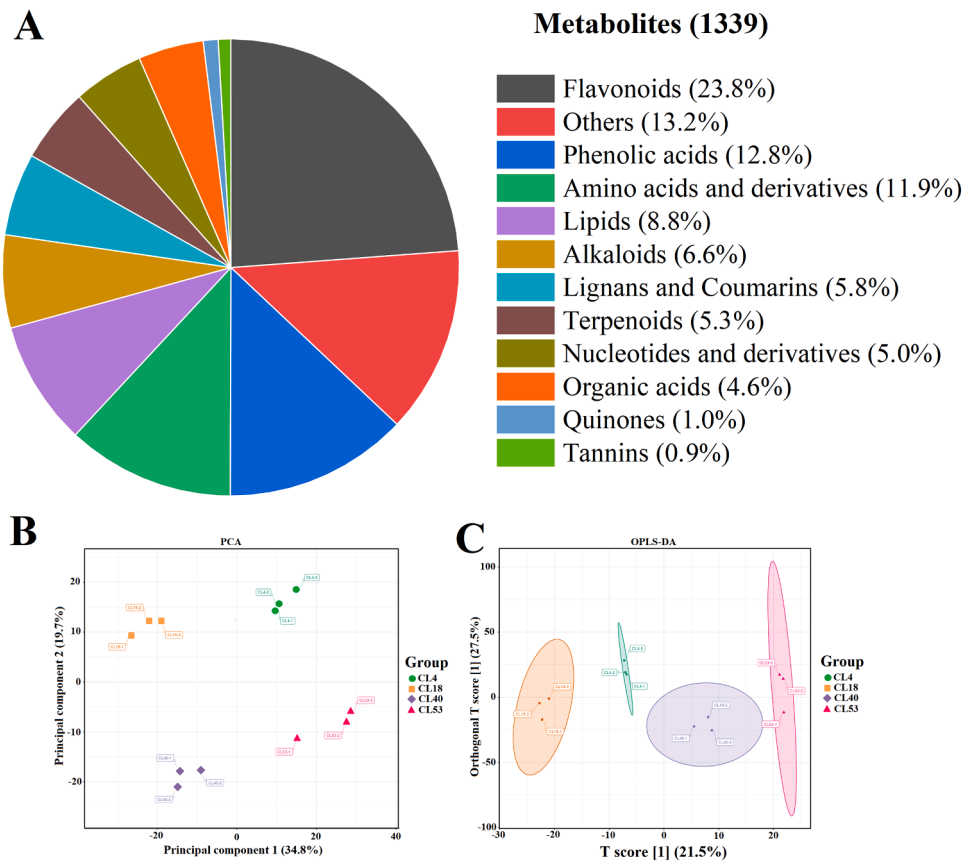


Fig. 2. Classification of the identified metabolites in *Camellia oleifera* seeds (A); PCA (B) and OPLS-DA (C) analyses of metabolites identified in *Camellia oleifera* seeds of four cultivars.

distinct clusters corresponding to cultivars, underscoring significant differences in their metabolite compositions. These findings strongly suggest that genetic variation plays a pivotal role in shaping the metabolite profiles of *Camellia oleifera* seeds.

As shown in Fig. 3A, 260 significantly differential metabolites were identified between CL18 and CL4, with 31 up-regulated and 229 down-regulated. Among them, protocatechuic acid xylosyl xyloside was the up-regulated metabolite with largest FC value (50.25), while chlorogenic acid methyl ester was the down-regulated metabolite with largest FC value (99.77). Between CL40 and CL4, 266 significantly differential metabolites were identified, with 54 up-regulated and 212 down-regulated (Fig. 3B). Among them, mudanpioside E showed the highest up-regulated FC value (44.62), while methyl caffeate was the down-regulated metabolite with largest FC value (87.57). In the comparison between CL53 and CL4, 117 significantly differential metabolites were identified, with 25 up-regulated and 92 down-regulated (Fig. 3C). The up-regulated metabolite with the highest FC value was 3-cinnamoylquinic acid (47.70), while 3-(3,4,5-Trimethoxyphenyl)propan-1-ol exhibited highest down-regulated FC values (83.93). There were 151 significantly differential metabolites between CL40 and CL18, with 81 up-regulated and 70 down-regulated (Fig. 3D). Among them, licorice glycoside C2 was the up-regulated metabolite with largest FC value (67.89), while syringaresinol showed the highest down-regulated FC value (35.98). In the comparison between CL53 and CL18, 165 significantly differential metabolites were identified, with 94 up-regulated and 71 down-regulated (Fig. 3E). The up-regulated metabolite with the highest FC value was 1-O-cinnamoyl-β-D-glucose (165.45), and 8R-dihydrodehydrodiconferyl alcohol 4-O-β-D-glucopyranoside exhibited the highest down-regulated FC value (105.92). Between CL53 and CL40, 153 significantly differential metabolites were identified, with 82 up-regulated and 71 down-regulated (Fig. 3F). Among them, rhodiosin was the up-regulated metabolite with largest FC value (101.18), while 8R-dihydrodehydrodiconferyl alcohol 4-O-β-D-glucopyranoside showed the highest down-regulated FC value (59.65).

The differential metabolites were counted according to their classification. As shown in Fig. 3G, the differential metabolites were categorized into 13 classes, of which, flavonoids and phenolic acids were the main specialized metabolites that were significantly altered among cultivars, while lipids, amino acids and derivatives were the primary metabolites that were significantly varied. Compared with CL4, most flavonoids and phenolic acids were down-regulated in the other three cultivars.

3.5.1. *Differences in flavonoids*

Flavonoids are important specialized metabolites in plants, displaying antioxidant, anti-inflammatory, antibacterial and anticancer activities (Kumar et al., 2013). A total of 319 flavonoid metabolites (97 flavonols, 82 flavones, 38 flavanones, 21 chalcones, 16 flavanols, 16 anthocyannins, 15 isoflavones, 9 flavanonols, 1 aurones, 1 dihydroisoflavones and 23 other flavonoids) were identified in *Camellia oleifera* seeds (Table S1). Among these, the most abundant were five flavonols (kaempferol-4′-O-glucoside, kaempferol-3-O-mannoside, kaempferol-3-O- glucorhamnoside, kaempferol-3-O-neohesperidoside and kaempferol-3-O-rhamnosyl (1→2)glucoside) and two flavones (luteolin-7-O-rutinoside and luteolin-7-O- neohesperidoside), whereas tricin-4′-O- (guaiacylglycerol)ether-7-O-glucoside, quercetin-3-O-glucoside-7-O-xyloside, luteolin-5-(3″,5″-O-Di-Rhamnosyl)glucoside -4′-glucoside and catechin-3-O-α-L-rhamnopyranoside were minor flavonoids present in *Camellia oleifera* seeds. Notably, 42 kaempferol derivatives were identified, constituting 45.28 % of total flavonoids, while 15 luteolin derivatives accounted for 19.43 %. In addition, quercetin derivatives (23 compounds) and apigenin derivatives (19 compounds)

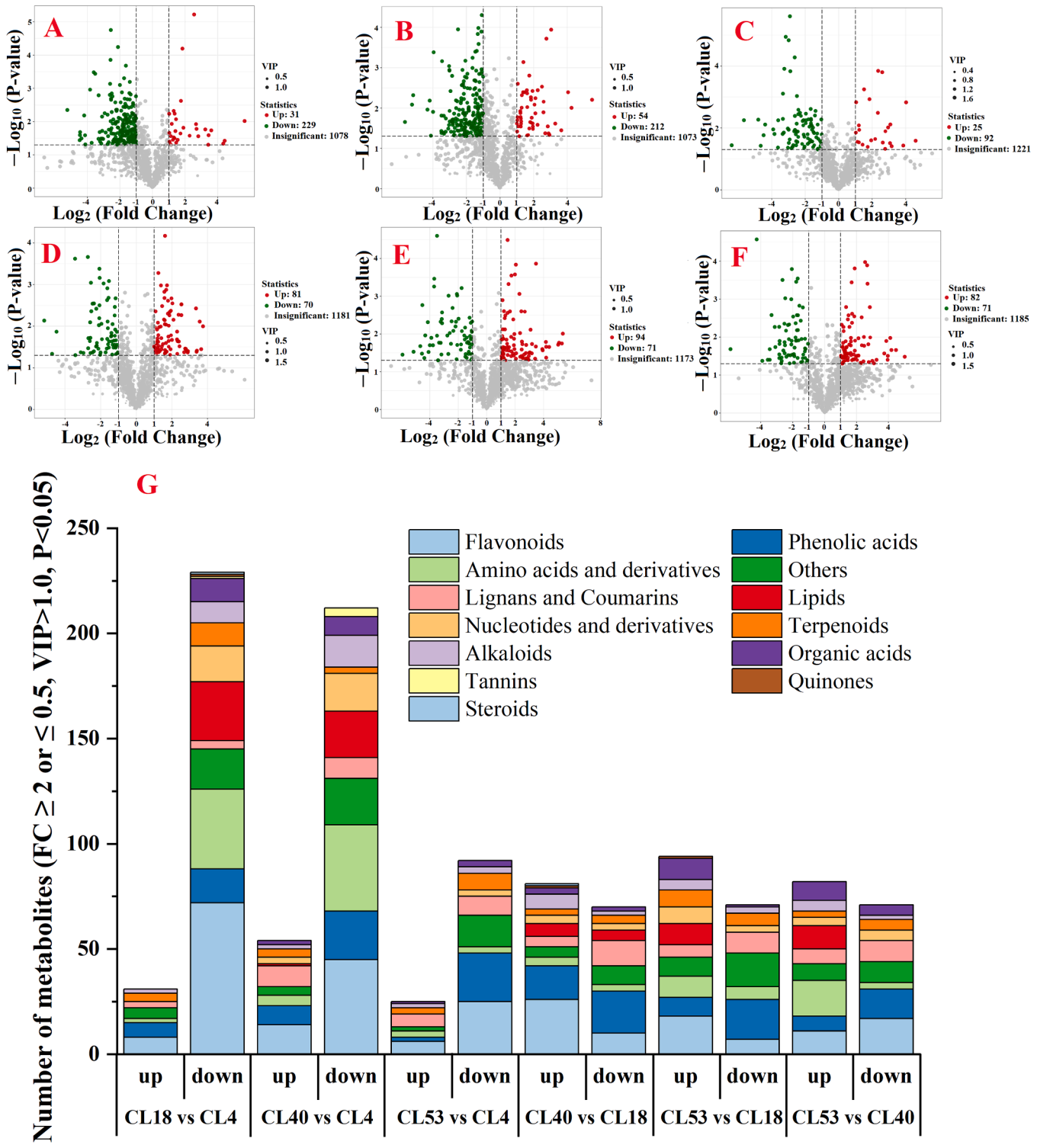


Fig. 3. Up-regulated and down-regulated metabolites between the six comparison groups. (A) CL18 vs CL4; (B) CL40 vs CL4; (C) CL53 vs CL4; (D) CL40 vs CL18; (E) CL53 vs CL18; (F) CL53 vs CL40.

were also the main flavonoid compounds.

Significant differences in 63 flavonoids (16 flavones, 15 flavonols, 11 flavanones, 3 chalcones, 3 isoflavones, 2 flavanols, 2 anthocyannins, 2 flavanonols, 1 aurones, 1 dihydroisoflavones and 7 other flavonoids) were observed among the four cultivars (Fig. 4A). CL53 exhibited higher levels of nearly half the significantly differential flavonoids, aligning with its higher TFC than other cultivars ($p < 0.05$). In CL18, eight flavonoids (e.g., quercetin-3-O-(2″-O-xylosyl)rutinoside, apigenin-6-C-glucoside) were more abundant than in CL4, CL40, and CL53. CL40 showed higher levels of acacetin-7-O-neohesperidoside, acacetin-7-O-rutinoside, taxifolin and tectochrysin than other cultivars. The levels of quercetin-3-O-arabinoside, isoluteolin, and isorhamnetin-3-0-arabinoside in CL4 were higher than those in other cultivars.

In the CL18/CL4 comparison, eight significantly differential flavonoids were more abundant in CL18, while the levels of 76 flavonoids were lower. Among the flavonoids at higher levels in CL18 than in CL4,

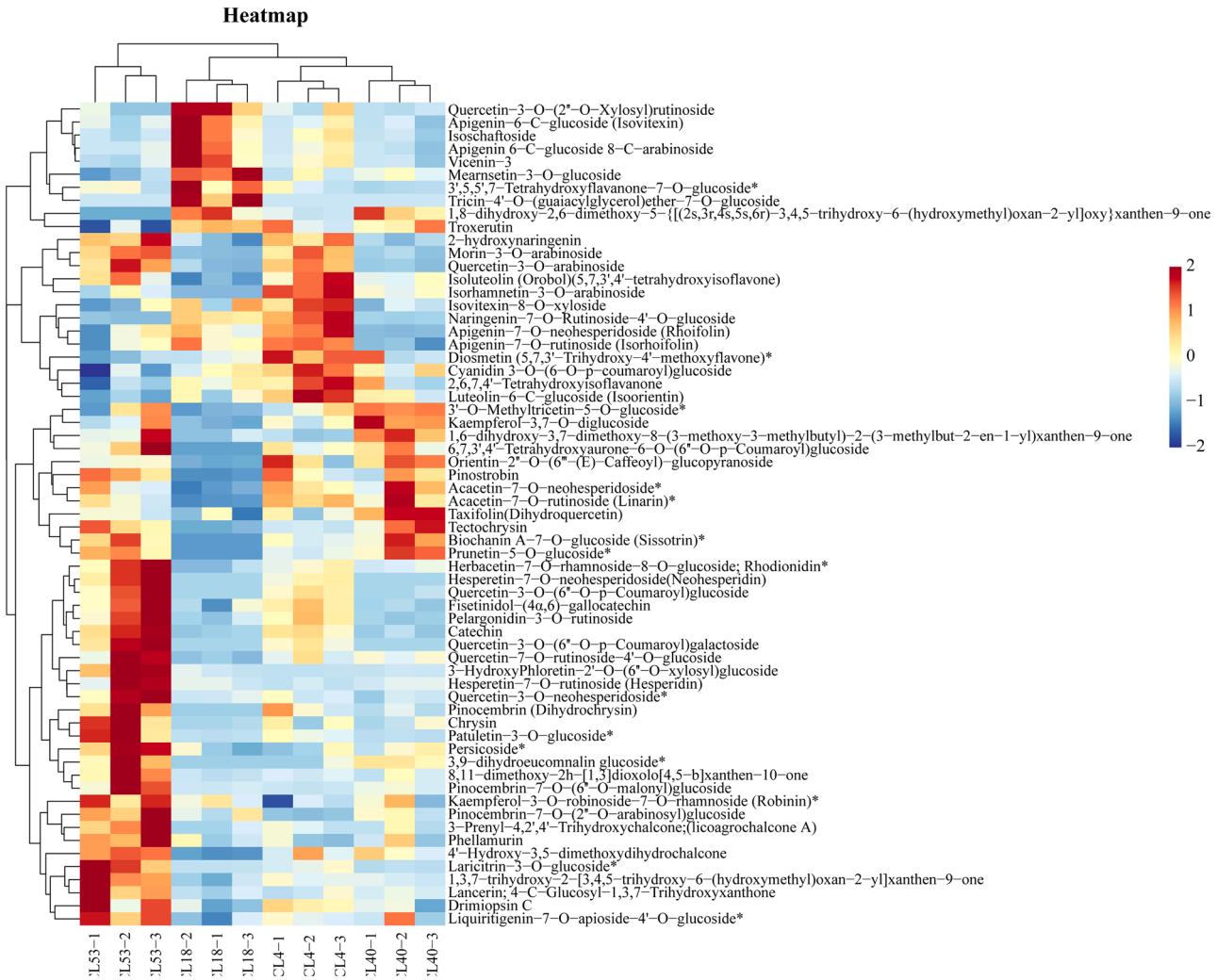


Fig. 4. Heatmap of significantly differential flavonoid metabolites (A) and phenolic acid metabolites (B) in *Camellia oleifera* seeds among four cultivars.

the compound with the largest FC was hispidulin-8-C-(2″-O-xylosyl) glucoside, whereas herbacetin-7-O-(3″-glucosyl) rhamnoside was found at 21.7-fold higher in CL4. The CL40/CL4 comparison revealed 14 significantly differential flavonoids with higher levels in CL40, and 45 flavonoids with lower levels. Six and 25 significantly differential flavonoids with higher and lower accumulation, respectively, were identified from the CL53/CL4 comparison. The CL40/CL18 comparison revealed 26 significantly differential flavonoids with higher levels in CL40, and 10 flavonoids with lower levels. 18 and seven significantly differential flavonoids with higher and lower accumulation, respectively, were identified from the CL53/CL18 comparison. In the CL53/CL40 comparison, 11 significantly differential flavonoids were more abundant in CL53, while 17 flavonoids were lower.

3.5.2. *Differences in phenolic acids*

Phenolic acids are important components of plants and plant-derived foods, and also possess numerous biological activities (Rashmi et al., 2020). In *Camellia oleifera* seeds, a total of 172 phenolic acid metabolites were identified, most of which were glycosidated with glucose (Table S1). Among them, cimicifugic acid E-glucose, 2,5-O-di-cinnamyl-sorbitol, 1-O-*p*-coumaroyl-β-D-glucose, 1-O-(*p*-coumaroyl) 3-hydroxy-3-methylglutaric acid and *p*-coumaric acid-4-O-glucoside were the most abundant phenolic acids, while 3,4-di-O-*p*-coumaroylquinic acid, 4-benzoyl-1,3,6-tri-O-β-D-galloylglucose and 3-dehydroshikimic acid-5-(3-vanilloyl) gallate were minor phenolic acids present in the four cultivars. Significant differences in 55 phenolic acids were observed among the four cultivars (Fig. 4B). In CL18, nearly half the significantly differential phenolic acids showed higher levels than other cultivars ($p < 0.05$), while most significantly differential phenolic acids in CL40 were present in lower levels than other cultivars.

The CL18/CL4 comparison identified seven and 16 phenolic acids with higher and lower accumulation, respectively. In particular, the abundance of protocatechuic acid xylosyl xyloside in CL18 was 50.25 times higher than in CL4, while the level of brevifolin carboxylic acid in CL4 was 14.20 times higher than in CL18. The CL40/CL4 comparison identified nine up-regulated and 22 down-regulated phenolic acids, with protocatechuic acid xylosyl xyloside showing the largest FC (19.11) and 5-galloyl-6-O-benzoyl glucose the minimum (0.0892). The CL53/CL4 comparison revealed two and 23 phenolic acids with higher and lower accumulation, respectively. 2-(caffeoyl) rhamnoside-O-rhamnoside and 2-(formylamino) benzoic acid were more abundant in CL53, while the level of 3-O-galloyl-D-glucose was significantly lower (FC: 0.0981) in CL53 than in CL4. In the CL40/CL18 comparison, 16 and 20 significantly differential phenolic acids with higher and lower accumulation, respectively, were identified. Among the up-regulated and down-regulated metabolites, 1′-O-galactoyl-*p*-coumaric acid and 3-O-galloyl-D-glucose showed the largest FC, respectively. The CL53/CL18 comparison revealed nine and 19 phenolic acids with higher and lower accumulation, respectively. 2-(caffeoyl) rhamnoside-O-rhamnoside was the up-regulated compound with the largest FC (40.93), and 4-O-glucosyl-sinapate was the down-regulated compound with the minimum FC (0.0477). The CL53/CL40 comparison identified seven up-regulated and

14 down-regulated phenolic acids, respectively. In particular, the abundance of methyl cinnamate in CL53 was 20.22 times higher than in CL40, while CL40 was noted to contain high level of brevifolin carboxylic acid.

3.6. KEGG annotation and enrichment analysis

The statistically significant differential metabolites were annotated using the KEGG database, and the annotation results were categorized according to KEGG pathway types. It was found that the differential metabolites among the four *Camellia oleifera* cultivars were mapped to 68 pathways, including "metabolic pathways", "biosynthesis of secondary metabolites", "carbon metabolism", "amino acid biosynthesis", "nucleotide biosynthesis", among others. Further KEGG pathway enrichment analysis was performed on the statistically significant differential metabolites of each comparison group, and the top 20 significantly enriched pathways were visualized using bubble plots (Fig. 5).

The differential metabolites of the CL18/CL4 comparison were involved in 80 pathways, with significant enrichment ($p < 0.05$) observed in "biosynthesis of quercetin aglycones II", "valine, leucine and isoleucine degradation", "nicotinate and nicotinamide metabolism", "phenylpropanoid biosynthesis", "nucleotide metabolism" and "propanoate metabolism" (Fig. 5A). Notably, these pathways were significantly down-regulated in CL18 compared with CL4. The metabolic pathway analysis of differential metabolites identified in the comparison of CL40/CL4 indicated that four significantly enriched pathways were "biosynthesis of quercetin aglycones II", "flavonoid biosynthesis", "caffeine metabolism" and "valine, leucine and isoleucine degradation" (Fig. 5B). The significantly enriched metabolic pathways in the comparison of CL53/CL4 were related to "stilbenoid, diarylheptanoid and gingerol biosynthesis" and "phenylpropanoid biosynthesis" (Fig. 5C). In the comparison of CL40/CL18, "flavonoid biosynthesis", "flavone and flavonol biosynthesis" and "phenylpropanoid biosynthesis" were significantly enriched metabolic pathways (Fig. 5D). In the comparison of CL53/CL18, the significantly enriched metabolic pathways included "biosynthesis of quercetin aglycones II", "linoleic acid metabolism", "pyruvate metabolism", "zeatin biosynthesis", and "propanoate metabolism" (Fig. 5E). In the comparison of CL53/CL40, significant enrichment was observed in 13 pathways, including "biosynthesis of quercetin aglycones II", "biosynthesis of amino acids", "flavonoid biosynthesis", among others (Fig. 5F). In summary, the metabolic pathways among the four *Camellia oleifera* cultivars were predominantly enriched in "biosynthesis of quercetin aglycones II" and "flavonoid biosynthesis", highlighting their potential roles in the metabolic differences.

3.7. Antioxidant activities of Camellia oleifera seeds

Estimation of antioxidant capacity using various assays is crucial for understanding overall antioxidant potential of food products (Wang et al., 2025). Fig. 6 showed the antioxidant activities of *Camellia oleifera* seeds against ABTS, DPPH and hydroxyl radicals, as well as FRAP. Among the four samples, CL53 exhibited the highest ABTS RSA (23.24 ± 1.35 %), followed by CL18 (21.72 ± 1.11 %), both of which surpassed CL40 (18.51 ± 0.98 %) and CL4 (18.01 ± 0.49 %) (Fig. 6A). A similar trend was observed for DPPH RSA, with CL53 showing the highest activity (66.13 ± 2.87 %), followed by CL18 (55.15 ± 3.11 %), CL40 (49.57 ± 1.83 %), and CL4 (48.90 ± 2.49 %) (Fig. 6B). As shown in Fig. 6C, the hydroxyl radical scavenging ability of the *Camellia oleifera* seed samples was significantly higher than their ABTS and DPPH scavenging capacities. The trend in hydroxyl RSA across the four cultivars was consistent with the results of the ABTS and DPPH assays. Additionally, the reducing capacity of CL53 (72.60 ± 2.89 μmol trolox/g DW) was significantly higher than those of CL18 (51.19 ± 2.55 μmol trolox/g DW), CL4 (36.94 ± 1.46 μmol trolox/g DW) and CL40 (36.85 ± 1.38 μmol trolox/g DW) (Fig. 6D). Overall, CL53 demonstrated the strongest antioxidant capacity, particularly in FRAP and DPPH free radical scavenging activities. The DPPH free radical scavenging rate of *Camellia oleifera* seed was comparable to the values of four well-characterized oilseeds (sesame, soybean, peanut and perilla) found by Kefale et al. (2023), in the case of its extract concentration much lower than those of other oilseeds (equivalent to one twentieth). Thus, *Camellia oleifera* seed may exhibit much higher antioxidant activity compared with common oilseeds. Furthermore, Spearman's correlation analysis was employed to reveal the relationship between the antioxidant activities and phenolic compounds. The TPC and TFC were noted to be positively correlated with the antioxidant activities (correlation coefficients: 0.89~0.99, $p < 0.05$), highlighting the significant contribution of these compounds to the antioxidant potential of *Camellia oleifera* seeds.

4. Discussion

Camellia oleifera seeds have traditionally been used to develop high-quality edible oil, which is gaining increasing attention (Cheng et al.,

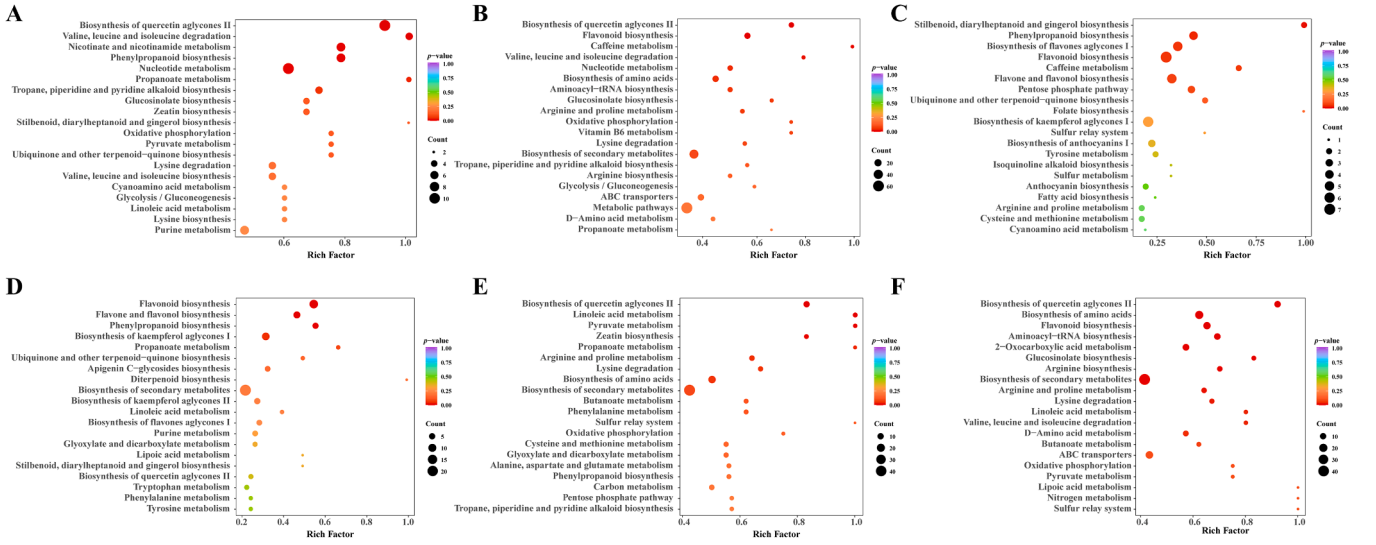


Fig. 5. KEGG enrichment of differential metabolites between the six comparison groups. (A) CL18 vs CL4; (B) CL40 vs CL4; (C) CL53 vs CL4; (D) CL40 vs CL18; (E) CL53 vs CL18; (F) CL53 vs CL40.

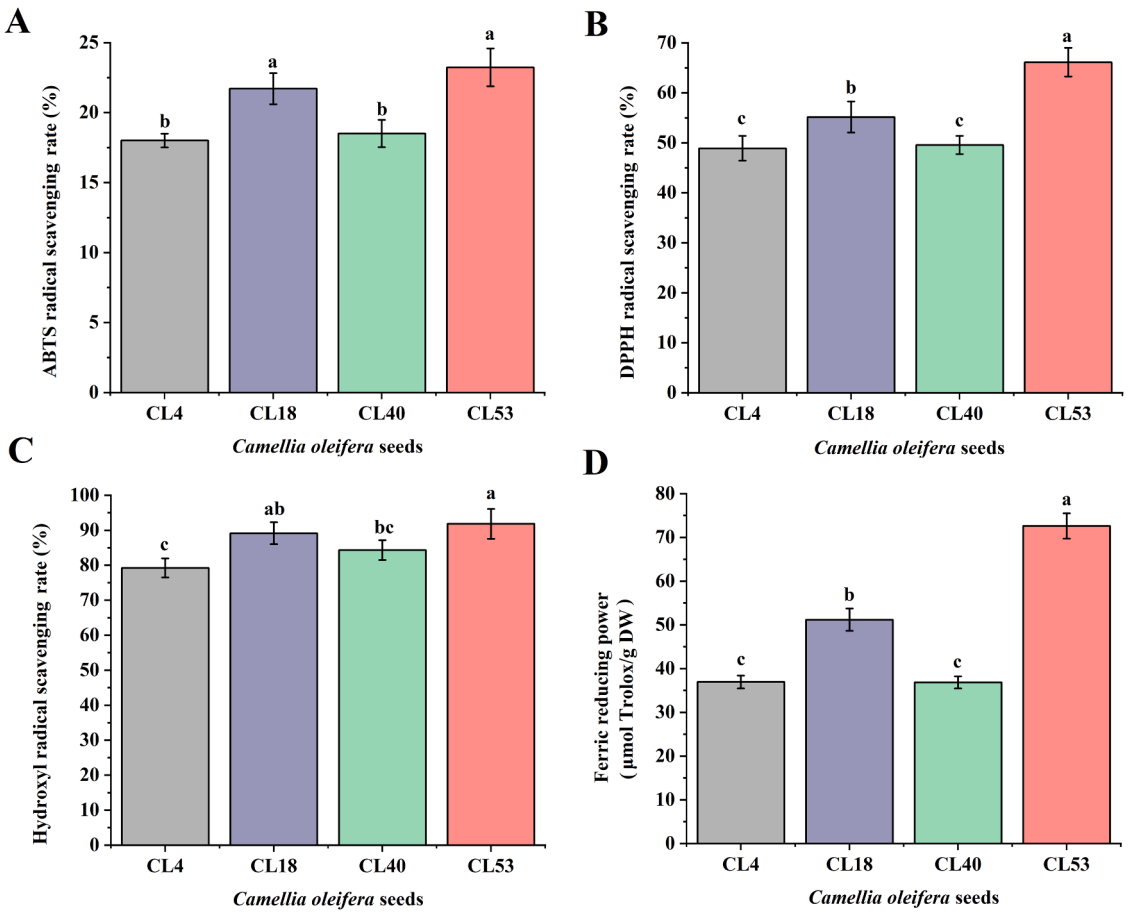


Fig. 6. Antioxidant properties of *Camellia oleifera* seeds. (A) ABTS radical scavenging ability; (B) DPPH radical scavenging ability; (C) hydroxyl radical scavenging ability; (D) FRAP reducing antioxidant power. Different small letters in a column indicate significant differences between samples ($p < 0.05$).

2015). However, beyond their use in oil production, the potential of *Camellia oleifera* seeds remains underexplored, as their functional active ingredients have not been thoroughly investigated. To date, only a limited number of compounds in *Camellia oleifera* seeds have been studied. Four kaempferol glycosides were identified in the flavonoid fraction of *Camellia oleifera* seeds, collectively accounting for 98.59 % of the total flavonoids, highlighting their dominance as the major flavonoids in these seeds (Liu et al., 2014). Hong et al. (2019) analyzed the polyphenols in the *Camellia oleifera* seed cakes and 73 phenolic compounds were identified, including flavonoid glycosides, phenolic acids and their derivatives, anthocyanins, and flavanones. Similarly, Song et al. (2022) explored the composition, distribution and molecular mechanisms of flavonoid biosynthesis in the *Camellia oleifera* seeds, and 148 flavonoid compounds were identified, including flavonols, flavones, flavanols, flavanones, flavanonols, among others. More recently, Xie et al. (2023) identified the composition profile of 44 polyphenols in *Camellia oleifera* seed cake, including hydroxybenzoic acids, hydroxycinnamic acids, flavonoids, flavanols, flavonols and flavanones.

In this study, we conducted a widely targeted metabolome analysis of *Camellia oleifera* seeds from four cultivars, providing a comprehensive metabolic profile of this valuable woody oil crop and significantly expanding the understanding of specialized metabolites in its seeds. Flavonoids and phenolic acids were the predominant specialized metabolites in *Camellia oleifera* seeds. These findings aligned with previous studies that reported similar kaempferol derivatives in *Camellia oleifera* seeds and their byproducts (Liu et al., 2014; Song et al., 2022; Xie et al., 2023). Among the flavonoids, kaempferol glycoside derivatives were the most abundant. Similarlly, Hong et al. (2019) reported that two flavonoids named kaempferol 3-[2G-glucosylrutinoside] and kaempferol 3-[2G-xylosylrutinoside] displayed the highest abundant peaks in the extracts of *Camellia oleifera* seed cakes. Additionally, luteolin glycosides, quercetin derivatives and apigenin glycosides were also identified as major flavonoid compounds. These highly accumulated flavonoids are known for their beneficial bioactivities. Kaempferol glycosides are naturally occurring phytochemicals with promising bioactivities, such as antioxidant, anti-inflammatory, antimicrobial, anticancer, neuroprotective, and hepatoprotective activities. They can reduce the lipid oxidation in the human body, prevent the organs and cell structure from deterioration and protect their functional integrity (Bangar et al., 2022). Among phenolic acids, compounds such as cimicifugic acid E-glucose, 2, 5-O-di-cinnamyl-sorbitol, 1-O-*p*-coumaroyl-β-D-glucose, 1-O-(*p*-coumaroyl) 3-hydroxy-3-methylglutaric acid and *p*-coumaric acid-4-O-glucoside were detected with significantly high levels. Most of the phenolic acids were glycosidated with glucose in this study, while *p*-coumaric acid was reported as the predominant compound in *Camellia oleifera* seed cakes by Hong et al. (2019). The identification of metabolites provides a valuable reference for the exploration of functional compounds from *Camellia oleifera* seeds and their by-products. The remaining material after oil extraction represents a cost-effective and renewable resource with significant potential for the development of high-value products (Qiu et al., 2020). However, as reported by Nicoli et al. (1999), post-harvest processing influenced the bioactive compounds of fruits and vegetables. Processing or prolonged storage can promote the enzymatic or chemical oxidation of phenolic compounds; while prolonged storage also can lead to hydrolysis or polymerization of phenolic compounds. Before utilizing the by-products of *Camellia oleifera* seeds, processes such as drying, storage and oil extraction are necessary, which will have an impact on the metabolites. Xia et al.

(2025) investigated the potential of *Camellia oleifera* seed residues as sources of bioenergy and explored the chemical richness of seed residues. This study laid the foundation for further exploration and utilization of processed *Camellia oleifera* seeds or residues to unlock their full functional and economic potential.

This study revealed 330 significantly differential metabolites among the four *Camellia oleifera* cultivars, and the results indicated remarkable differences in their metabolite profiles. Flavonoids and phenolic acids were the main specialized metabolites that that exhibited significant variation, and amino acids and derivatives were the primary metabolites that significantly varied. The levels of flavonoids exhibited significant difference across the four cultivars, suggesting that the accumulation of certain compounds in the flavonoid biosynthesis pathways was cultivar-dependent. In addition, these differences in flavonoids composition and content may contribute to the variation in antioxidant activities observed among the cultivars. Similarly, the levels of phenolic acids also showed significant differences, which might be ascribed to a combination of genetic factors, soil conditions, and environmental influences that affect the quality and quantity of phytochemicals. These findings aligned with previous reports that some phenolics varied among different *Camellia oleifera* cultivars (Hong et al., 2019) and were consistent with HCA results, which revealed substantial differences in metabolite profiles among the cultivars. The up- and down-regulated metabolites with the greatest difference among the pairwise cultivar comparisons were basically phenolic compounds (5 phenolic acids, 2 flavonoids and 3 lignans). Increasing evidence has shown that consumption of a variety of phenolic compounds lower the risk of health disorders because of their numerous biological activities.

Metabolic pathway analysis revealed that the differential metabolites among the four *Camellia oleifera* cultivars were mainly mapped to "metabolic pathways", "biosynthesis of secondary metabolites", "carbon metabolism", "amino acid biosynthesis", and "nucleotide biosynthesis". Further KEGG pathway enrichment analysis highlighted significant enrichment in pathways related to "biosynthesis of quercetin aglycones II" and "flavonoid biosynthesis", indicating that the differential metabolite profiles of the four *Camellia oleifera* cultivars were closely associated with the specialized metabolites. Further comparative studies on the regulation of these pathways in different *Camellia oleifera* cultivars by utilizing multi-omics approaches (metabolomics, transcriptomics, lipomics and proteomics) are needed for identifying key candidate genes or genetic markers for this woody oilseed's quality improvement.

Generally, phenolics, including flavonoids and phenolic acids, are well-known for their strong antioxidant properties and diverse biological and pharmacological activities (El-Saadony et al., 2024; Shen et al., 2022). Phenolic compounds demonstrate promising effects in the pharmaceutical applications, as for the prevention of various diseases such as diabetes, obesity, cancer, cardiovascular diseases, osteoporosis, neurodegenerative diseases, and also for dermatological therapy. Due to their antioxidant effects, phenolic compounds can be used for food preservation and bioactive packaging, as natural dyes, elaboration of hydrogels and nanocomplexes. Therefore, these phenolic compounds have been widely applied in pharmacy, food chemistry, materials engineering and fine chemistry (de Araujo et al., 2021). It's worth noting that the health effects of phenolic compounds depend on their bioavailability. Metabolism of phenolic compounds *in vivo* occurs via a common pathway with the aglycones absorbed directly from the small intestine. However, most phenolic compounds are present in the form of glycosides, esters, or polymers that cannot be absorbed in their native form. These substances must be hydrolyzed by intestinal enzymes or by the colonic microflora before being absorbed (Manach et al., 2004). In *Camellia oleifera* seeds, most flavonoids and phenolic acids were found in glycosylated or esterified forms, which probably resist acid hydrolysis in the stomach and thus arrive intact in the colon. Research on utilization of gut microbiota is an urgent need to improve the efficacy and bioavailability of the abundant kaempferol, luteolin, quercetin and phenolic acid glycosides identified in *Camellia oleifera* seeds.

Presumably, the high levels of the phenolic compounds contribute to the enhanced antioxidant capacity observed in the *Camellia oleifera* seeds (Zhang, et al., 2023). Specifically, the superior overall antioxidant activity of CL53 may be correlated with its higher levels of flavonoids and phenolic acids, which are often integrated with polysaccharides in plant cell walls. Kefale et al. (2023) investigated the metabolic profiles of four oilseeds (sesame, soybean, peanut, and perilla), and revealed that the differences of bioactive flavonoids and phenolic acids led to the variations in antioxidant activity. Therefore, the variation in antioxidant activities among *Camellia oleifera* cultivars can be attributed to differences in the levels of phenolic compounds. The identification of these differential metabolites provides valuable insights into the functional and nutritional evaluation of different *Camellia oleifera* cultivars, highlighting their potential for developing high-value products with enhanced health benefits.

5. Conclusions

In this study, a UHPLC-QqQ-MS/MS-based widely targeted metabolomics approach was employed to identify and compare the metabolites in the seeds of four *Camellia oleifera* cultivars. The analysis revealed comprehensive insights into the composition and abundance of metabolites, with a total of 1339 metabolites categorized into 13 distinct classes. Significant variation in metabolite profiles was observed among different cultivars, particularly in the accumulation of flavonoids and phenolic acids, which might lead to differences in their health-promoting properties. Metabolic pathway analysis of differential metabolites highlighted significant enrichment in the "biosynthesis of quercetin aglycones II" and "flavonoid biosynthesis" pathway. Comparison of phenolic compounds (including flavonoids and phenolic acids) and antioxidant capacity across the cultivars further demonstrated that these compounds are key contributors to the observed differences in antioxidant activities. This study provides the firsthand comprehensive information on the metabolites and bioactive compounds with health-promoting functions in *Camellia oleifera* seeds. The findings offer a scientific foundation for selecting target cultivars with enhanced functional properties, thereby supporting the development of value-added applications for this economically important woody oil crop.

Ethics

This study did not involve human or animal subjects, and thus, no ethical approval was required. The study protocol adhered to the guidelines established by the journal.

CRediT authorship contribution statement

Zuoyi Zhu: Writing – original draft, Validation, Methodology. **Jingfang Lv:** Investigation, Data curation. **Jiarong Yao:** Validation, Formal analysis. **Zhen Li:** Writing – review & editing, Formal analysis. **Shuai Yang:** Methodology, Data curation. **Xueping Zhao:** Supervision, Conceptualization. **Fen Dai:** Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.afres.2026.101700.

Data availability

Data will be made available on request.

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